

HOT SPRINGS, AR, USA

WMO: 723415

Lat: 34.478N

Lon: 93.096W

Elev: 535

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	25.1	5.4	7.4	27.9	10.0	9.3	31.7	17.9	49.2	16.3	49.1	4.0	80	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	98.6	75.4	96.0	75.7	93.2	75.4	79.3	90.4	78.4	89.4	77.6	88.5	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	141.0	82.9	75.4	135.8	82.7	74.7	132.4	82.4	43.0	90.7	42.2	89.7	41.3	88.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.4	12.1	14.7	102.1	3.6	4.0	12.1	105.0	10.0	107.3	8.0	109.6	5.4	112.6		
			12.9	80.7	3.3	1.2	10.5	81.6	8.6	82.3	6.8	83.0	4.4	83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	42.6	45.6	54.6	63.3	71.0	79.4	82.6	82.1	75.5	64.0	53.2	45.3
	DBStd	16.18	9.79	9.99	9.32	7.64	6.63	4.39	4.39	4.70	6.00	8.29	8.92	8.86
	HDD50	796	271	188	58	6	1	0	0	0	0	3	66	203
	HDD65	2831	696	545	340	119	27	0	0	0	3	122	366	613
	CDD50	5672	41	66	202	405	651	883	1010	994	765	437	162	56
	CDD65	2233	2	2	19	68	212	433	545	529	319	91	12	1
	CDH74	23281	2	12	122	552	1763	4589	6303	6024	3072	783	57	2
	CDH80	10512	0	0	11	90	537	2095	3185	3041	1315	234	4	0
Wind	WSAvg	5.1	5.4	5.9	6.2	6.1	5.2	4.7	4.3	4.3	4.4	4.6	5.0	5.1
Precipitation	PrecAvg	54.8	4.1	3.7	5.3	5.4	6.3	4.1	4.4	2.8	4.0	5.0	4.9	5.0
	PrecMax	82.2	9.4	8.0	10.3	10.6	15.6	10.2	11.5	8.7	11.2	14.7	11.7	10.1
	PrecMin	37.1	0.6	1.1	1.4	2.2	0.6	1.3	0.2	0.5	0.4	0.4	1.4	0.6
	PrecStd	10.6	2.2	1.5	2.4	2.6	3.4	2.3	2.3	1.9	2.4	3.3	2.7	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	75.6	81.4	85.6	90.6	98.9	102.0	102.2	97.5	90.1	80.2	72.6
		MCWB	62.0	62.6	64.3	67.7	72.1	74.2	75.1	76.4	73.8	69.8	66.9	63.7
	2%	DB	67.9	70.7	77.4	82.2	88.1	95.0	98.5	98.9	93.7	86.0	74.8	68.3
		MCWB	59.3	61.0	62.6	66.4	71.4	75.2	75.5	75.7	73.5	70.5	63.7	62.2
	5%	DB	64.1	66.4	73.4	80.4	85.6	92.2	95.5	96.1	90.8	82.0	71.9	64.4
		MCWB	57.9	58.6	60.6	65.5	71.2	74.7	76.3	76.0	73.3	68.0	63.3	58.4
	10%	DB	60.3	63.1	70.3	77.2	82.5	90.1	92.8	92.8	87.9	78.5	68.2	60.9
		MCWB	51.4	55.3	59.3	64.0	70.2	74.3	76.3	75.7	72.5	66.6	60.4	54.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	67.0	68.0	72.8	76.5	79.5	80.5	80.4	77.9	75.4	71.0	67.1
		MCDB	69.4	70.9	77.2	81.0	84.6	90.5	92.0	90.7	88.1	84.0	76.0	70.4
	2%	WB	63.1	64.2	65.8	70.2	74.9	78.1	79.3	79.3	76.6	73.5	67.7	64.0
		MCDB	66.0	68.2	74.0	77.9	82.9	88.9	90.6	90.3	86.8	81.2	72.9	67.4
	5%	WB	59.4	60.4	63.8	68.7	73.5	77.1	78.4	78.3	75.6	71.6	65.0	60.1
		MCDB	62.9	65.2	70.5	76.2	81.8	87.6	89.4	89.5	85.1	78.4	69.7	63.5
	10%	WB	52.7	56.7	61.6	67.0	72.2	76.0	77.7	77.4	74.6	69.4	61.9	55.7
		MCDB	57.1	61.1	68.5	74.4	80.2	86.0	88.5	88.6	83.5	75.1	67.0	59.2
Mean Daily Temperature Range	5% DB	MDBR	19.1	18.6	20.7	21.0	18.8	18.7	19.0	19.2	20.3	21.1	20.0	18.5
		MCDBR	24.2	23.6	23.9	23.4	21.3	21.6	22.4	23.1	23.5	24.4	22.7	22.4
		MCWBR	17.0	15.9	12.9	10.9	8.3	6.7	5.7	6.1	7.2	10.9	13.7	16.4
	5% WB	MCDBR	18.2	19.1	19.5	17.7	17.4	18.0	18.9	19.5	18.5	18.8	17.6	18.7
		MCWBR	16.9	15.7	12.9	10.4	8.5	6.9	6.2	6.5	6.7	10.2	14.0	16.1
Clear-Sky Solar Irradiance	taub	0.323	0.327	0.356	0.391	0.423	0.460	0.478	0.476	0.405	0.357	0.334	0.312	
	taud	2.497	2.486	2.428	2.332	2.304	2.224	2.211	2.199	2.379	2.488	2.505	2.540	
	Ebn at Noon	281	292	291	284	275	263	258	256	272	278	274	279	
	Edh at Noon	27	30	35	40	42	45	45	45	36	30	26	24	
All-Sky Solar Radiation	RadAvg	801	935	1303	1635	1779	2004	1960	1807	1554	1206	860	700	
	RadStd	72	135	107	102	156	175	117	138	126	171	100	82	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+1.62	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page